

Descriptive & Inferential Statistics

Summarising data, and inferring beyond the sample

PROFESSIONAL DEFINITION

Statistics fall into two families. Descriptive statistics summarise and describe the data at hand — central tendency (mean, median, mode), dispersion (range, standard deviation) and distribution. Inferential statistics use sample data to draw conclusions about a wider population — hypothesis tests, confidence intervals, correlation and regression. The test chosen must match the data type (nominal, ordinal, interval/ratio) and the question asked; a mismatched test invalidates the result (Field, 2018).

DESCRIPTIVE

Describe what the data shows. Central tendency: mean, median, mode. Dispersion: range, variance, standard deviation. Shape: skew, kurtosis. Visuals: histograms, box plots, bar charts. Always the first step — know your data before testing it.

INFERENTIAL

Generalise beyond the sample. t-tests (compare two means), ANOVA (three or more), chi-square (categorical association), correlation (strength of relationship), regression (prediction). Report the test, statistic, degrees of freedom, p-value and effect size.

Significance is not importance. A p-value below 0.05 says an effect is unlikely to be chance — not that it is large or meaningful. Always report an effect size alongside significance, and interpret it in business terms. A statistically significant but trivially small effect is a common over-claim.

Question	Data type	Test
Compare 2 group means	Interval	t-test
Compare 3+ means	Interval	ANOVA
Association of categories	Nominal	Chi-square
Strength of relationship	Interval	Correlation
Predict an outcome	Interval	Regression

CHECK ASSUMPTIONS

Parametric tests assume normality, homogeneity of variance and adequate n. Check them and report; use non-parametric tests if violated.

REPORT FULLY

State the test, statistic, df, p-value and effect size. 'It was significant' with no numbers is not reportable analysis.

MATCH TEST TO DATA

Ordinal data does not take a mean; categorical data does not take a t-test. The wrong test invalidates the conclusion.

Common pitfalls. (1) Reporting significance without effect size. (2) A test that does not match the data type. (3) Ignoring assumption checks. (4) Confusing correlation with causation. (5) Over-interpreting a tiny but 'significant' effect.

REFERENCES

Field, A. (2018) *Discovering Statistics Using IBM SPSS Statistics*. 5th edn. London: SAGE. · Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research Methods for Business Students*. 8th edn. Harlow: Pearson.